

Customer Shopping Behavior Analysis

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.





Dataset Summary

3,900

Total Purchases

Rows of transactional data analyzed

18

Columns

Features across demographics, purchases, and behavior

37

Missing Values

In the Review Rating column, imputed
by category median

Key features include customer demographics (Age, Gender, Location, Subscription Status), purchase details (Item, Category, Amount, Season, Size, Color), and shopping behavior (Discount Applied, Previous Purchases, Frequency, Review Rating, Shipping Type).

Exploratory Data Analysis in Python

We began with data preparation and cleaning in Python, importing the dataset using `pandas` and exploring structure with `df.info()` and `.describe()` for summary statistics.

01

Missing Data Handling

Imputed missing Review Rating values using the median rating of each product category.

03

Feature Engineering

Created `age_group` by binning ages and `purchase_frequency_days` from purchase data.

02

Column Standardization

Renamed columns to snake case for better readability and documentation.

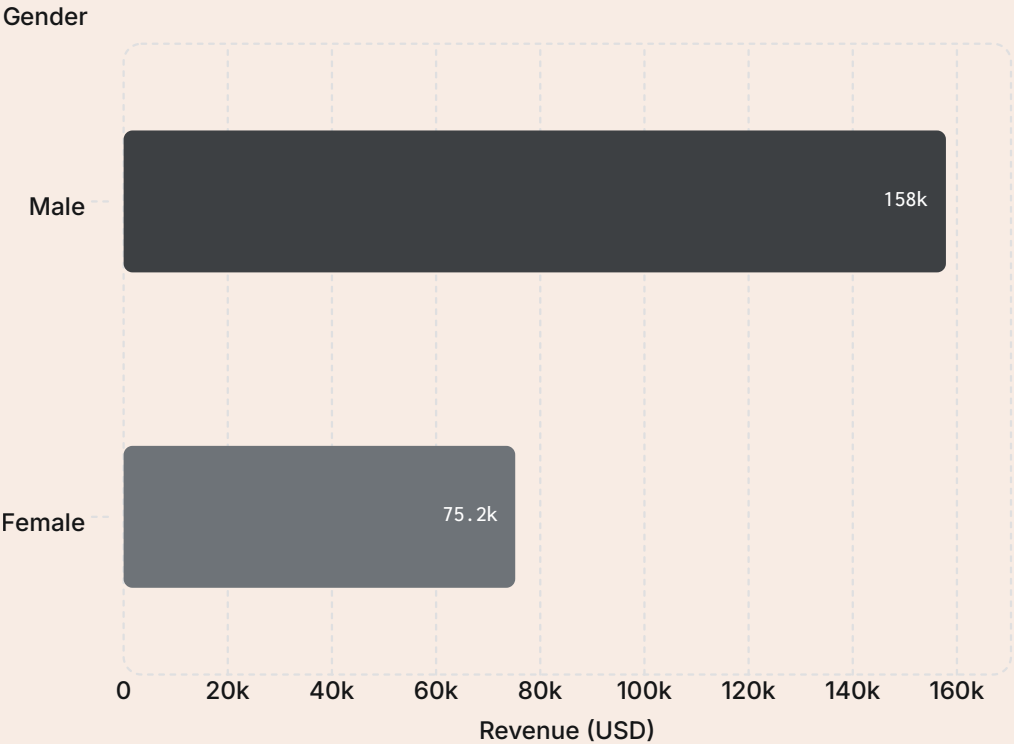
04

Database Integration

Loaded the cleaned DataFrame into PostgreSQL for structured SQL analysis.

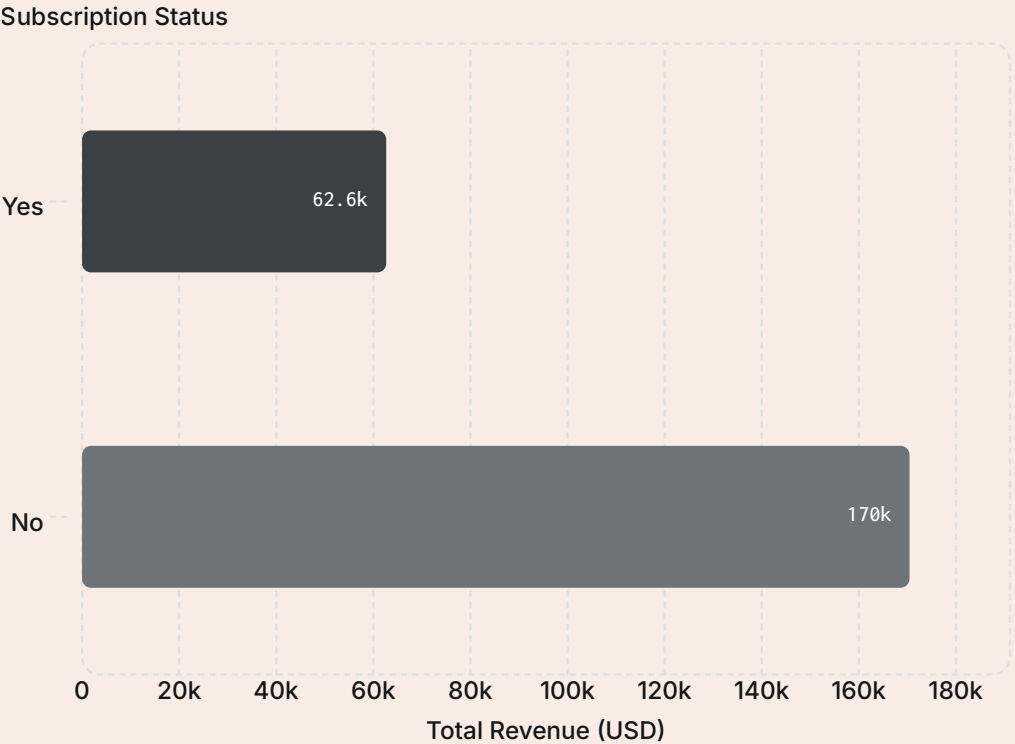
Revenue by Gender & Subscription Status

Revenue by Gender



Male customers generated \$157,890 vs. \$75,191 from female customers – more than double the revenue.

Subscribers vs. Non-Subscribers



Non-subscribers drove \$170,436 in revenue across 2,847 customers, while 1,053 subscribers contributed \$62,645. Average spend was nearly identical at ~\$59.50-\$59.87.

Top-Rated Products & Discount-Dependent Items

Top 5 Products by Rating

Product	Avg Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

Top 5 Discount-Dependent Products

Product	Discount Rate (%)
Hat	50.00
Sneakers	49.66
Coat	49.07
Sweater	48.17
Pants	47.37

Top-rated products like Gloves and Sandals earn customer satisfaction, while Hats and Sneakers lead in discount dependency – nearly half their purchases involve discounts.

Customer Segmentation & Shipping

Loyal Customers

3,116 customers – the dominant segment, representing strong repeat business.

Returning

701 customers who have made repeat purchases but haven't reached loyal status.

New

83 customers – a small but important group for growth and onboarding efforts.

Shipping Type Comparison

Shipping Type	Avg Purchase (USD)
Standard	\$58.46
Express	\$60.48

Express shipping customers spend slightly more on average than Standard shipping users.

Repeat Buyers & Subscriptions

Subscribed?	Repeat Buyers
No	2,518
Yes	958

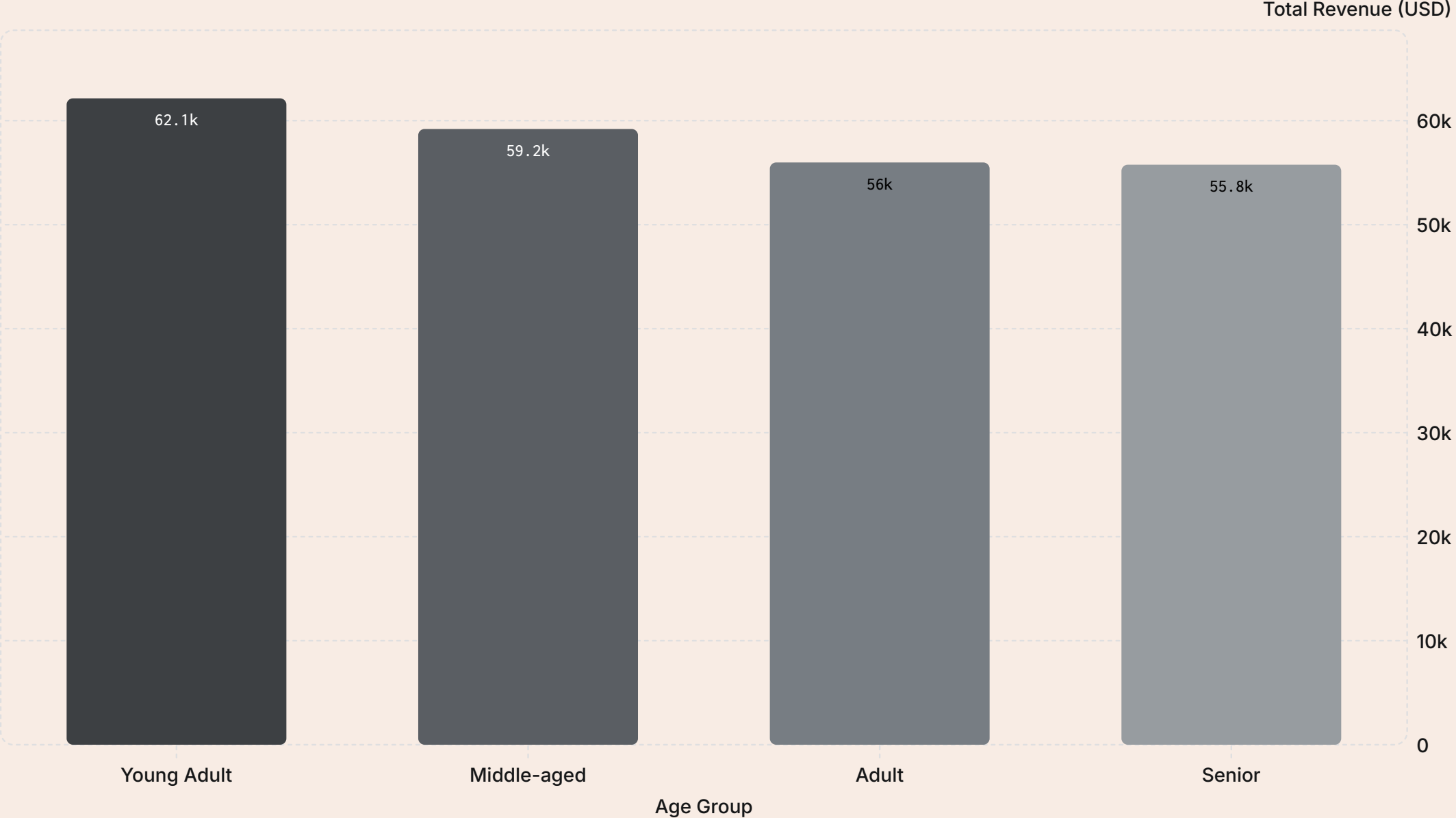
Among customers with more than 5 purchases, most remain non-subscribers – a clear opportunity.



Top Products per Category

1	Accessories Jewelry (171), Sunglasses (161), Belt (161)
2	Clothing Blouse (171), Pants (171), Shirt (169)
3	Footwear Sandals (160), Shoes (150), Sneakers (145)
4	Outerwear Jacket (163), Coat (161)

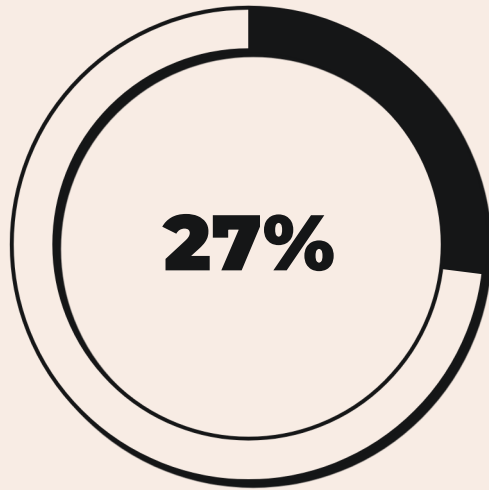
Revenue by Age Group



Young Adults lead revenue generation at \$62,143, followed closely by Middle-aged customers at \$59,197. Adults and Seniors contribute \$55,978 and \$55,763 respectively – a relatively even distribution across age groups.

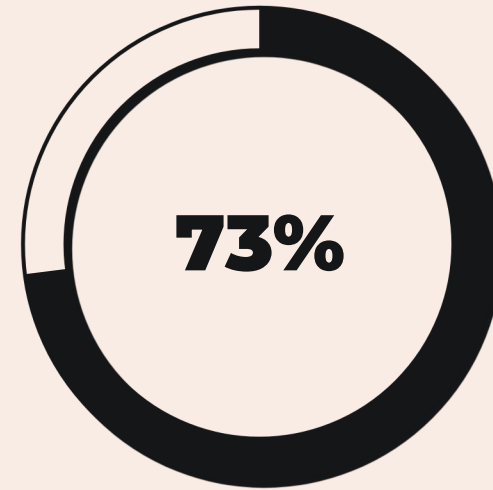
Power BI Dashboard

We built an interactive dashboard in Power BI to present insights visually. Key metrics include 3.9K customers, \$59.76 average purchase amount, and a 3.75 average review rating.



Subscribers

Of total customer base



Non-Subscribers

Of total customer base

The dashboard features interactive filters for subscription status, gender, category, and shipping type, enabling stakeholders to explore revenue by category and age group dynamically.

Business Recommendations



Boost Subscriptions

Promote exclusive benefits for subscribers to convert the 73% non-subscriber base.



Loyalty Programs

Reward repeat buyers to move them into the Loyal segment of 3,116 customers.



Review Discount Policy

Balance sales boosts with margin control – nearly half of Hats and Sneakers sell on discount.



Targeted Marketing

Focus on high-revenue age groups like Young Adults and express-shipping users.